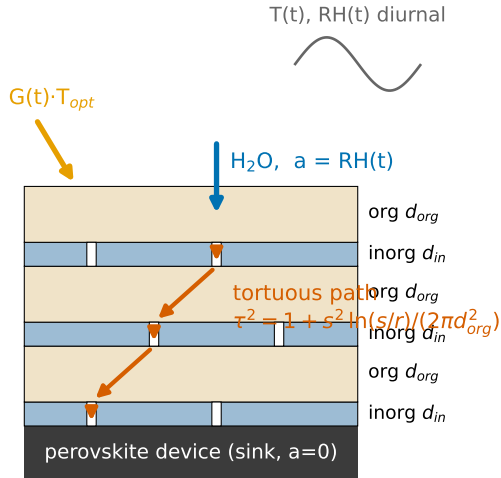


(a) Stack, defects and forcing



(b) Governing model

Moisture (activity form): $S(x) \partial_t a = \partial_x (P(x, T) \partial_x a)$

$P_{in}^{eff} = P_{lat}(T) + f(d) P_{org}(T)$ (pinhole parallel path)

$$f(d) = f_0 e^{-d/d_c} + f_{res} + f_{c0} \left(\frac{d - d_{crit}}{d_{crit}} \right)_+^2$$

$$P_{org}^{eff} = P_{org}/\tau^2, \quad s = r\sqrt{\pi/f} \quad (\text{lateral decoupling})$$

Heat: $\rho c_p \partial_t T = \partial_x (k \partial_x T) + q_{solar}$; Arrhenius $D(T)$

Time lag (exact, layer-lumped Frisch): $t_{lag} = \sum_j C_j R_{L,j} R_{R,j} / R_{tot}$

$$T80 = t_{lag} + M_{crit} R_{tot} / \Delta a$$

Numerics: FV, harmonic conductance, implicit Euler — 0.45% vs analytic